

AY25 PROGRAM ASSESSMENT REPORT

v2026.1

Tracking Data

CIM code (e.g., ME-CM)	MAS-CYBS-1
Program Name	Master of Cybersecurity
Assessment Year	Spring 25
Program Assessment Coordinator	Yousef Elmehdwi
Year of Current Assessment Plan	2025
No Assessment This Year per Plan	☐
LO(s) Assessed	G2: Students will demonstrate their familiarity with the goals, main theoretical constructs, key algorithmic methods, and know how to apply them to cybersecurity aspects pertaining to software security.
Presented at program authority meeting date: <i>Please attach minutes</i>	

Methodology

This report is based on student work collected from Introduction to Information Security, which was offered during this assessment cycle. Although both Introduction to Information Security and Software Security are aligned with Learning Outcome G2, the Software Security course was not offered that term due to lack of instructor and is scheduled for Spring 2026. Therefore, only Introduction to Information Security was used for assessment this cycle.

- Assignments and exams assess students' understanding of fundamental principles in software and information security, including encryption systems, operating system and database security, network security, system threats, and risk avoidance procedures.

This rubric categorizes students into three assessment levels: *Does not meet expectations (D)*, *Meets expectations (M)*, and *Proficient (P)*, based on their performance on key information security concepts, including theoretical understanding, algorithmic methods, and application to cybersecurity scenarios.

Criteria	Weight	How Measured	Level of Understanding	Assessment Category Description	Percent of Students Achieving Measure
Understanding of Information Security Principles & Application of Security Algorithms	100%	Assignments and Exams	Below 60%: (D)	Shows little or no understanding of core InfoSec concepts, including encryption, OS security, database security, network security, system threats, risk avoidance, and application of security algorithms.	17%
			60% - 79%: (M)	Demonstrates basic understanding; able to explain concepts and apply algorithms with minor errors or gaps in understanding.	33%
			P: 80%-100%	Strong understanding; applies concepts and algorithms accurately and effectively in assignments.	33%
			N: Not assessed	Withdrawn from the course.	17%

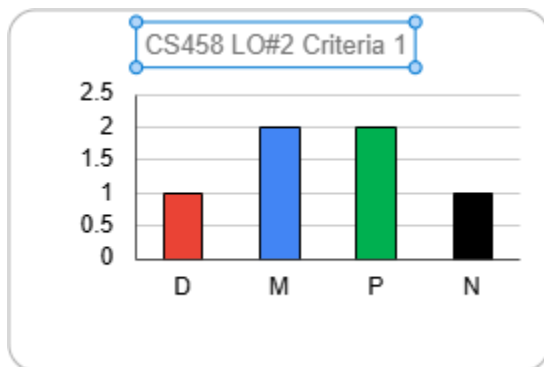
Results

Artifact samples

See attached file.

Data table

A#	major	NAME	student	major	D/M/P	comment for N
A0000001	MAS-CYBS-1		student 1	MAS-CYBS-1	M	
A0000002	MAS-CYBS-1		student 2	MAS-CYBS-1	M	
A0000003	MAS-CYBS-1		student 3	MAS-CYBS-1	P	
A0000004	MAS-CYBS-1		student 4	MAS-CYBS-1	D	
A0000005	MAS-CYBS-1		student 5	MAS-CYBS-1	N	Withdrew
A0000006	MAS-CYBS-1		student 6	MAS-CYBS-1	P	



Description of findings

Continuous Improvement, CI, plans

In Spring 2025, only six Cybersecurity students were enrolled in the course, while most of the class (over 100 students) came from other CS programs/specializations at both the undergraduate and graduate levels.

One-third of students demonstrated proficiency (80–100%), while another third achieved only a basic understanding (60–79%). A smaller portion (17%) scored below expectations, indicating difficulty with fundamental cybersecurity concepts. Additionally, 17% were not assessed due to withdrawal.

Areas of Concern:

- Many students are in the basic range and need more support to build deeper understanding.
- Few students reached proficiency, showing a need for stronger teaching strategies.

Note:

- The core course CS527 Software Security was scheduled for assessment this cycle but was not offered. It will be offered in Spring '26 and assessed then.

Prior CI plan impact

☒ no previous CI plans available.

This cycle's CI plan

☐ After consideration and presentation to the program authority, no CI plans are recommended.

To address these concerns, the following actions will be implemented:

1. Frequent Quizzes and Assessments
 - Short quizzes after each major topic (e.g., encryption, OS security, network security) will be scheduled every two weeks with immediate feedback to check understanding.
2. Hands-On Exercises and Class Project:
 - Increase practical exercises during lectures, integrate applied learning into the class project, and adapt SEED Labs activities to provide structured, guided practice.
3. Encouraging Office Hours Attendance
 - Promote office hours as a resource for personalized support, announce them in class, and track attendance to follow up with students needing extra help.
4. Assessment of CI Plan Effectiveness Plan:
 - Evaluate the success of these strategies by reviewing quiz scores, project results, and student feedback at the end of the cycle to guide adjustments for future terms.

Implementation

- Changes will be implemented before Fall 2026, with ongoing monitoring during the semester.
- Course instructors will develop and deliver instructional improvements.
- The program coordinator will oversee implementation and evaluate the effectiveness of these CI measures.